

习题13.1 p172 1(1,4,6), 2,4,5

习题13.2 p179 1-4中任选3道题, 5-8中任选3道题, 共6道题

习题13.3 p189 1,3, 6-9中任选2道题

习题13.5 p203 奇数中任选3道题, 偶数中任选3道题, 共6道题

习题13.6 p208 所有12道题中中任选6道题

13.1

13.1.1

Determine radius of convergence, and convergence at boundaries.

(1) $\sum \frac{x^n}{n(n+1)}$: $\rho = 1$, convergent at $x = \pm 1$;

(4) $\sum H_n x^n$: $\rho = 1$, diverges at $x = \pm 1$.

(6) $\sum \frac{(\log n)^\alpha}{n^\alpha} x^n$, $\alpha \in \mathbb{R}$: $\rho = 1$, converges at $x = 1$ iff $\alpha > 1$, converges at $x = -1$ iff $\alpha > 0$.

13.1.2

Determine radius of convergence of $\sum \frac{\sin n\alpha}{n} x^n$.

Solution: $\limsup_{n \rightarrow \infty} (\sin n\alpha)^{1/n} = 1$ so $\rho = 1$.

13.1.4

Suppose $l = \liminf_{n \rightarrow \infty} |a_n/a_{n+1}|$, $L = \limsup_{n \rightarrow \infty} |a_n/a_{n+1}|$, and the radius of convergence of $\sum a_n x^n$ is R , prove that $L^{-1} \leq R \leq l^{-1}$.

Proof: Trivial since

$$\liminf_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \leq \liminf_{n \rightarrow \infty} |a_n|^{1/n} \leq \limsup_{n \rightarrow \infty} |a_n|^{1/n} \leq \limsup_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|.$$

13.1.5

Suppose $\sum a_n x^n$ has radius of convergence 1, and $\lim_{x \rightarrow 1^-} \sum a_n x^n = A$. If $a_n \geq 0$, show that $A = \sum a_n$.

Proof: Clearly $\sum_{n \leq N} a_n = \lim_{x \rightarrow 1^-} \sum_{n \leq N} a_n x^n \leq \lim_{x \rightarrow 1^-} \sum a_n x^n = A$ and $A = \lim_{x \rightarrow 1^-} \sum a_n x^n \leq \sum a_n$. Hence $A = \sum a_n$.

13.2

13.2.3

Suppose $S(x) = \sum a_n x^n$ has radius of converges $R > 0$, prove that

(1) If $\sum \frac{a_n}{n+1} R^{n+1}$ converges then $\int_0^R S(x) dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1} R^{n+1}$.

(2) If $\sum n a_n R^{n-1}$ converges, then $S'(R) = \sum n a_n R^{n-1}$.

Proof: (1) By Dominated Convergence Theorem

$$\int_0^R S(x) dx = \lim_{N \rightarrow \infty} \int_0^R \sum_{n=0}^N a_n x^n dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1} R^{n+1}.$$

(2) Then $g(x) = \sum n a_n x^{n-1}$ is continuous on $[-R, R]$, so $\frac{S(R)-S(x)}{R-x} = S'(\xi) = g(\xi)$ for some $\xi \in (x, R)$. Let $x \rightarrow R$, then $\lim_{x \rightarrow R} g(x) = S'(R)$ so $S'(R) = \sum n a_n R^{n-1}$.

13.2.4

Suppose $\sum a_n x^n$ converges uniformly on $(-R, R)$. Show that it converges uniformly on $[-R, R]$.

Proof: For any ε there exists N such that $n > m > N$ implies $|\sum_{k=n}^m a_k x^k| < \varepsilon \forall x \in (-R, R)$. Since $\sum_{k=n}^m a_k x^k$ is a polynomial, it is continuous hence $|\sum_{k=n}^m a_k x^k| \leq \varepsilon \forall x \in [-R, R]$, so $\sum a_n x^n$ converges uniformly on $[-R, R]$.

13.2.5

(Du Bois Reymond) Suppose $f \in C([0, 1])$ and $\int_0^1 f(x) x^n dx = 0 \forall n \geq 1$, then $f \equiv 0$.

Proof: For any polynomial $P \in \mathbb{R}[x]$, $\int_0^1 f(x) P(x) dx = 0$. Since $f \in C([0, 1])$ can be approximated by polynomials uniformly, $\int_0^1 f(x)^2 dx = 0$ so $f \equiv 0$.

13.2.6

Suppose a nonconstant f can be expanded as a power series at every $x \in (a, b)$, show that $f^{-1}(0)$ has no limit points in (a, b) .

Proof: Suppose $f(c) = 0$ and $f(x_n) = 0$ where $x_n \rightarrow c$. Then if $f(x) = \sum a_n (x - c)^n$, and a_k is the first non-zero term, for small h we have $|f(c + h)| \geq |a_k| \frac{h^k}{2}$ so f does not vanish in a small neighborhood of c , a contradiction. Hence the power series is identically zero, and $f \equiv 0$ in some neighborhood of c . Therefore the component of $f^{-1}(0)$ containing c is both closed and open, and it is non-empty, so $f^{-1}(0) = (a, b)$ a contradiction.

13.2.7

Suppose f can be expanded as a power series in some neighborhood of $x = 0$, and $f^{(n)}(0)$ is bounded. Show that f is the restriction of some $g \in C^\infty(\mathbb{R})$.

Proof: Let $g(x) = \sum \frac{f^{(n)}(0)}{n!} x^n$ then $g \in C^\infty(\mathbb{R})$. Since $f^{(n)}(0)$ is bounded, $R_n(x) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k$ converges to 0 uniformly in any compact set, so $f \equiv g$ locally. By Ex13.2.6 f is the restriction of g to its domain.

13.2.8

Prove that the Bernstein polynomial converges uniformly to f for $f \in C([0, 1])$.

Proof: Note that $\sum_{k=0}^n f(x) \binom{n}{k} x^k (1-x)^{n-k} = f(x)$, and $B_{f,n} = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}$. Let $p_{n,k}(x) = \binom{n}{k} x^k (1-x)^{n-k}$, then $\sum_{0 \leq k \leq n} p_{n,k}(x) \left(x - \frac{k}{n}\right)^2 = \frac{x(1-x)}{n}$.

Hence

$$|f(x) - B_{f,n}(x)| \leq \sum_{k=0}^n \left| f(x) - f\left(\frac{k}{n}\right) \right| p_{n,k}(x).$$

Let

$$\begin{aligned} \text{I} &= \sum_{|m-nx| < n^{3/4}} \left(f(x) - f\left(\frac{m}{n}\right) \right) p_{n,m}(x), \\ \text{II} &= \sum_{|m-nx| > n^{3/4}} \left(f(x) - f\left(\frac{m}{n}\right) \right) p_{n,m}(x). \end{aligned}$$

Then $|f - B_{f,n}| \leq |\text{I}| + |\text{II}|$.

For any $\varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall x \in [0, 1], n \geq N \implies |\text{I}| < \varepsilon$, since

$$|\text{I}| \leq \sup_{|x-m/n| < n^{-1/4}} |f(x) - f(m/n)| \rightarrow 0.$$

Suppose $M = \sup_{x \in [0,1]} |f(x)|$, then

$$|\text{II}| \leq 2M \sum_{|m-nx| > n^{3/4}} p_{n,m}(x) \leq 2M\sqrt{n} \sum_{m=0}^n (x - m/n)^2 p_{n,m}(x) = \frac{2Mx(1-x)}{\sqrt{n}}.$$

Hence $\|f - B_{f,n}\|_{\infty} \rightarrow 0$.

13.3

13.3.1

Calculate $\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+1}}{4n+1}$.

Solution:

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+1}}{4n+1} = \frac{\sqrt{2}}{4} \arctan\left(\frac{x\sqrt{2}}{1-x^2}\right) + \frac{\sqrt{2}}{8} \ln\left(\frac{x^2 + x\sqrt{2} + 1}{x^2 - x\sqrt{2} + 1}\right)$$

13.3.6

Calculate

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{3k+2}.$$

Solution: Note that

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{3k+2} = \sum_{k=0}^{\infty} \frac{1}{3k+1} - \frac{1}{3k+2} = L(1, \chi_3).$$

where $\chi_3 : (\mathbb{Z}/3\mathbb{Z})^\times \hookrightarrow \mathbb{C}^*$. Recall for odd χ ,

$$L(1, \chi) = \frac{\pi}{2q} \sum_{k=0}^{q-1} \chi(k) \cot \frac{k\pi}{q} = \frac{\pi}{6} \left(\cot \frac{\pi}{3} - \cot \frac{2\pi}{3} \right) = \frac{2\sqrt{3}}{9} \pi.$$

13.3.7

Calculate

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{4k+1}.$$

Solution: $(\mathbb{Z}/8)^\times = \{1, 3, 5, 7\}$. Consider $\chi : (1, 3, 5, 7) \mapsto (1, -1, -1, 1)$ and $\chi' : (1, 3, 5, 7) \mapsto (1, 1, -1, -1)$ then

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{4k+1} = \frac{1}{2} (L(1, \chi) + L(1, \chi')).$$

and likewise calculate $L(1, \chi') = \frac{\sqrt{2}\pi}{4}$, $L(1, \chi) = -\frac{\tau(\chi)}{q} \sum_{k=1}^{q-1} \log \sin \frac{k\pi}{q} = \frac{\sqrt{2}}{2} \log(1 + \sqrt{2})$. Therefore

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{4k+1} = \frac{\sqrt{2}\pi}{8} + \frac{\sqrt{2}}{4} \log(1 + \sqrt{2}).$$

13.4
